

Ayush Srihari

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EDUCATION

University of British Columbia

Expected graduation: May 2027

B.Sc. in Computer Science – **Dean's List 1x**

Vancouver, BC

Relevant Coursework: Machine Learning, Intro to AI, Networks, Architecture, Software Eng, Databases.

EXPERIENCE

AWS Cloud Innovation Centre @ UBC

Vancouver, BC

Software Developer Intern

September 2025 – Present

- Led **end-to-end development** of open-source GenAI applications for 3+ sponsor teams, demonstrating project ownership by facilitating stakeholder meetings and mentoring juniors on clean code.
- Optimized serverless environments by **lazy loading** heavy Python dependencies and decoupling core logic, minimizing container image sizes and initialization overhead.
- Engineered a **pre-warming strategy** utilizing **Provisioned Concurrency** and connection pooling, reducing Lambda cold starts from 2 mins to **sub-second** and cutting database costs by **60%**.
- Automated deployments via a standardized **AWS CI/CD pipeline** (CodeBuild, CodePipeline, ECR) with **multi-stage Docker builds**, reducing deployment latency by **90%** and eliminating configuration errors.
- Represented the CIC at **BCNET Connect 2026**, delivering technical workshops and project showcases to **800+ EdTech professionals** and provincial stakeholders.

Coast Capital Savings

Vancouver, BC

Project Manager & Developer

January 2025 – May 2025

- Led a team of **8 developers** to deliver a full-stack document management platform for mortgage applications, managing sprint planning, code reviews, and stakeholder communication.
- Architected a serverless **OCR pipeline** using AWS Textract and OpenCV to extract structured fields and barcodes from multi-format documents (PDF, Image, DOCX).
- Built a React + TypeScript frontend with JWT-based session management via **AWS Cognito** and role-based access control to secure sensitive financial data.
- Deployed 10+ AWS Lambda functions behind **API Gateway**, storing metadata in Aurora Serverless RDS and raw files across dedicated S3 buckets.

PROJECTS

GenRx – AI Simulation Platform

AWS CDK, Bedrock, LangChain, ECS Fargate, pgvector

March 2026

- Architected a multi-tenant AI platform for 4+ clinical departments, managing 5,000+ vector embeddings in pgvector with **sub-100ms retrieval latency** via LangChain RAG pipelines.
- Developed a seamless **voice-to-voice interface** on ECS Fargate using WebSockets and Amazon Nova Sonic 2, achieving **sub-200ms** round-trip latency for bidirectional simulations.
- Engineered an **automated pedagogical debriefing** module that assesses student-patient interactions against clinical rubrics, generating instant grading and feedback via LLM evaluation.

OpenED – AI Study Companion

AWS CDK, Bedrock, Glue, Lambda, LangChain, RAG, Docker

Dec 2025

- Developed a cloud-native AI companion sponsored by BCcampus, enabling **100,000+** users to streamline learning using AWS CDK, Lambda, and Bedrock with an Amplify frontend.
- Built scalable **ETL pipelines** using AWS Glue to process 200+ textbooks, generating embeddings via Cohere Embed v4 for **sub-second similarity retrieval** across 100,000+ chunks in pgvector.
- Orchestrated asynchronous knowledge retrieval and **WebSocket progress streaming** with semantic caching to minimize LLM API overhead and redundant generation costs.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C/C++, SQL

Cloud & Infrastructure: AWS (CDK, Lambda, ECS, Bedrock, Glue, RDS), Docker, CI/CD, Git

Frameworks & Tools: Node.js, React.js, Express, FastAPI, LangChain, PostgreSQL, MongoDB, OpenAPI